Lines & Angles

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

For the model $f(x) = 5x + 4$ in deductive reasoning, evaluate $f(5)$.

Final Answer: 29. Substitute $x = 5$: $f(5) = 5(5) + 4 = 29$.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In ΔABC studying properties of equality and congruence, $m\angle A = 39^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 96°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (39^\circ + 45^\circ) = 96^\circ$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

For the model $f(x) = 2x + 6$ in proving segment relationships, evaluate $f(2)$.

Final Answer: 10. Substitute $x = 2$: $f(2) = 2(2) + 6 = 10$.

4 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

A data set recorded for proving segment relationships consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 17.5. The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

5 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

In ΔABC studying proving angle relationships, $m\angle A = 41^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 94°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (41^\circ + 45^\circ) = 94^\circ$.

6 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Two parallel lines L_1 and L_2 are cut by a transversal T. A pair of alternate exterior angles are given by $(7x - 14)^\circ$ and $(4x + 31)^\circ$. Find the value of x and the measure of each alternate exterior angle.

Step 1: Set up the equation using alternate exterior angles: Alternate exterior angles created by parallel lines are congruent: $7x - 14 = 4x + 31 \Rightarrow 3x = 45 \Rightarrow x = 15$. Step 2: Evaluate the angle measure: $7(15) - 14 = 105 - 14 = 91^\circ$. Check: $4(15) + 31 = 60 + 31 = 91^\circ$. Both alternate exterior angles measure 91°.

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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Two parallel lines are intersected by a transversal line. Two consecutive interior angles have measures $(3x - 10)^\circ$ and $(2x + 25)^\circ$. Find the measure of the acute angle.

Solution: Consecutive interior angles are supplementary, so they sum to 180°: $(3x - 10) + (2x + 25) = 180 \Rightarrow 5x + 15 = 180 \Rightarrow 5x = 165 \Rightarrow x = 33$. Substitute $x = 33$ into both angle expressions: $3(33) - 10 = 89^\circ, \quad 2(33) + 25 = 91^\circ$. The acute angle has measure 89°.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Lines & Angles · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Two angles $\angle P$ and $\angle Q$ form a linear pair. If $m\angle P = (4x + 12)^\circ$ and $m\angle Q = (2x + 18)^\circ$, find the measure of the smaller angle.

Solution: Since they form a linear pair, their measures sum to 180°: $(4x + 12) + (2x + 18) = 180 \Rightarrow 6x + 30 = 180 \Rightarrow 6x = 150 \Rightarrow x = 25$. Now substitute $x = 25$ back into each expression: $m\angle P = 4(25) + 12 = 112^\circ, \quad m\angle Q = 2(25) + 18 = 68^\circ$. The smaller angle has measure 68°.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two angles $\angle 1$ and $\angle 2$ are vertical angles. If $m\angle 1 = (5x - 23)^\circ$ and $m\angle 2 = (3x + 17)^\circ$, find the measure of $m\angle 1$ and the measure of its supplement.

Step 1: Set up the equation using vertical angles: Since vertical angles are equal in measure: $5x - 23 = 3x + 17 \Rightarrow 2x = 40 \Rightarrow x = 20$. Step 2: Find the measure of $m\angle 1$: $m\angle 1 = 5(20) - 23 = 100 - 23 = 77^\circ$. Step 3: Find its supplement: $180^\circ - 77^\circ = 103^\circ$.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In ΔABC, the exterior angle at vertex C measures 134°. The two remote interior angles have measures $m\angle A = (5x - 4)^\circ$ and $m\angle B = (3x + 10)^\circ$. Find the measure of $\angle A$.

Solution: By the Exterior Angle Theorem, an exterior angle equals the sum of the two remote interior angles: $(5x - 4) + (3x + 10) = 134 \Rightarrow 8x + 6 = 134 \Rightarrow 8x = 128 \Rightarrow x = 16$. Substitute $x = 16$ into $m\angle A$: $m\angle A = 5(16) - 4 = 80 - 4 = 76^\circ$.

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12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In isosceles triangle ΔPQR, $PQ = PR$. The vertex angle $\angle P$ measures 40°. An angle bisector is drawn from base angle $\angle Q$ to point D on side PR. What is the measure of $\angle QDR$?

Step 1: Find the base angles of ΔPQR: Since $PQ = PR$, the base angles $\angle Q$ and $\angle R$ are equal: $m\angle Q = m\angle R = \frac{180^\circ - 40^\circ}{2} = \frac{140^\circ}{2} = 70^\circ$. Step 2: Find the bisected angle measure: QD bisects $\angle PQR$, so: $m\angle DQR = \frac{70^\circ}{2} = 35^\circ$. Step 3: Apply the triangle angle sum to ΔQDR.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

An equilateral triangle has side lengths of 12 cm. An altitude is drawn from one vertex to the opposite side. Find the exact height and area of the triangle.

Solution: The altitude bisects the vertex angle into two 30° angles and bisects the base into two equal segments of length 6 cm, forming two congruent 30–60–90 right triangles. The shorter leg opposite 30° is $x = 6$. The altitude (height) is the longer leg opposite 60°: $h = x\sqrt{3} = 6\sqrt{3}$ cm. Now compute the area using $\text{Area} = \frac{1}{2}bh$: $\text{Area} = \frac{1}{2}(12)(6\sqrt{3}) = 36\sqrt{3}$ cm^2.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Two similar cylinders have surface areas of 64π cm^2 and 144π cm^2. If the volume of the smaller cylinder is 96π cm^3, find the volume of the larger cylinder.

Step 1: Find the area ratio and linear scale factor k: $\frac{\text{Area}_2}{\text{Area}_1} = \frac{144\pi}{64\pi} = \frac{9}{4} = k^2 \Rightarrow k = \sqrt{\frac{9}{4}} = \frac{3}{2}$ Step 2: Compute the volume ratio k^3: $\frac{\text{Volume}_2}{\text{Volume}_1} = k^3 = \left(\frac{3}{2}\right)^3 = \frac{27}{8}$ Step 3: Calculate the volume of the larger cylinder: $\text{Volume}_2 = 96\pi \times \frac{27}{8} = 12\pi \times 27 = 324\pi$ cm^3

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Circles

Segment AB has endpoints $A(2, -1)$ and $B(6, 7)$. Find the equation of the perpendicular bisector of AB in slope-intercept form.

Solution: The perpendicular bisector passes through the midpoint and has a perpendicular slope. Step 1 (Midpoint): $M = \left(\frac{2+6}{2}, \frac{-1+7}{2}\right) = (4, 3)$. Step 2 (Slope of AB): $m_{AB} = \frac{7-(-1)}{6-2} = \frac{8}{4} = 2$. Step 3 (Perpendicular slope): $m_\perp = -\frac{1}{2}$. Step 4 (Point-slope equation through M(4, 3)): $y - 3 = -\frac{1}{2}(x - 4) \Rightarrow y - 3 = -\frac{1}{2}x + 2 \Rightarrow y = -\frac{1}{2}x + 5$.

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16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 8.01 Area, Surface Area & Volume**

In ΔPQR, ray QS bisects $\angle Q$ with S lying on side PR. If $PQ = 14$, $QR = 21$, and $PR = 25$, find the length of segment PS.

Step 1: Apply the Angle Bisector Theorem: The bisector divides side PR into segments proportional to the adjacent sides: $\frac{PS}{SR} = \frac{PQ}{QR} = \frac{14}{21} = \frac{2}{3}$ Step 2: Express SR in terms of PS: Since $PS + SR = 25$, we have $SR = 25 - PS$: $\frac{PS}{25 - PS} = \frac{2}{3} \Rightarrow 3PS = 2(25 - PS) = 50 - 2PS$ Step 3: Solve for PS: $5PS = 50 \Rightarrow PS = 10$ (and $SR = 15$).

17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 9.01 Transformations & Symmetry**

Find the coordinates of the point P on directed line segment from $A(-3, 2)$ to $B(7, 12)$ that partitions the segment in the ratio $2 : 3$.

Step 1: Determine the fractional distance: A ratio of $2 : 3$ means P is $\frac{2}{2+3} = \frac{2}{5}$ of the way from A to B. Step 2: Apply the partition formula: $x_P = x_1 + \frac{2}{5}(x_2 - x_1) = -3 + \frac{2}{5}(7 - (-3)) = -3 + \frac{2}{5}(10) = -3 + 4 = 1.$ $y_P = y_1 + \frac{2}{5}(y_2 - y_1) = 2 + \frac{2}{5}(12 - 2) = 2 + \frac{2}{5}(10) = 2 + 4 = 6.$ The partitioning point is $P(1, 6)$.

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 1.01 Foundations & Reasoning**

Triangle ΔABC has vertices $A(1, 2)$, $B(3, 2)$, and $C(1, 5)$. The triangle is dilated by a scale factor of $k = 3$ centered at the origin, and then translated by vector $\langle -2, 4 \rangle$. What are the coordinates of vertex C''?

Step 1: Dilation centered at the origin: Multiply each coordinate of $C(1, 5)$ by the scale factor $k = 3$: $C(1, 5) \longrightarrow C'(3 \times 1, 3 \times 5) = C'(3, 15)$. Step 2: Translation by vector $\langle -2, 4 \rangle$: Add -2 to the x-coordinate and 4 to the y-coordinate: $C'(3, 15) \longrightarrow C''(3 - 2, 15 + 4) = C''(1, 19)$. The final coordinates of vertex C'' are $(1, 19)$.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 10.01 Coordinate Geometry**

A solid metal sphere of radius 3 cm is melted down and recast into a cylinder of radius 2 cm. What is the height of the recast cylinder in cm?

Step 1: Calculate the volume of the sphere: $V_{\text{sphere}} = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(3^3) = \frac{4}{3}\pi(27) = 36\pi$ cm^3. Step 2: Equate with the volume of the cylinder: $V_{\text{cyl}} = \pi R^2 h = \pi(2^2)h = 4\pi h$. Step 3: Solve for height h: $4\pi h = 36\pi \Rightarrow h = \frac{36\pi}{4\pi} = 9$ cm. The height of the cylinder is 9 cm.

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20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 11.01 Probability

Point $A(-2, 4)$ is reflected across the line $y = x$, and the resulting point is then translated by the vector $\langle 3, -1 \rangle$. What are the final coordinates?

Solution: Apply the transformations step by step: Step 1: Reflection across $y = x$ swaps the coordinates: $(x, y) \to (y, x)$: $A(-2, 4) \longrightarrow A'(4, -2)$. Step 2: Translation by $\langle 3, -1 \rangle$ adds 3 to the x-coordinate and subtracts 1 from the y-coordinate: $A'(4, -2) \longrightarrow A''(4 + 3, -2 - 1) = A''(7, -3)$. The final coordinates are $(7, -3)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 9$ in conditional statements, evaluate $f(2)$.

a. 15 ✓

b. 18

c. 6

d. 19

Substitute $x = 2$: $f(2) = 3(2) + 9 = 15$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 3$ in language of logic, evaluate $f(2)$.

a. 11 ✓

b. 15

c. 8

d. 15

Substitute $x = 2$: $f(2) = 4(2) + 3 = 11$.

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23 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In the diagram below, lines m and n are parallel. A transversal line intersects m and n. Another line segment connects a point on m to a point on n, forming a "crook-angle". Given that $\angle 1 = 120^\circ$ and $\angle 2 = 135^\circ$, what is the measure of the "crook-angle" $\angle 3$?

a. 255°

b. **105° ✓**

c. 270°

d. 75°

To solve this problem, we construct an auxiliary line through the vertex of $\angle 3$ that is parallel to lines m and n. Let's call this auxiliary line p. This construction divides $\angle 3$ into two angles, let's call them $\angle 3a$ and $\angle 3b$, such that $\angle 3 = \angle 3a + \angle 3b$. Since line m is parallel to line p, and the transversal that forms $\angle 1$ intersects both m and p, $\angle 1$ and $\angle 3a$ are consecutive interior angles if we extend the transversal line. However.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two parallel lines, p and q, are intersected by a transversal line t. A pair of consecutive interior angles have measures $(4x + 9)^\circ$ and $(2x + 33)^\circ$. What is the value of x?

a. 8

b. 12

c. **23 ✓**

d. 37

By the Consecutive Interior Angles Theorem, when two parallel lines are cut by a transversal, consecutive interior angles are supplementary (their measures sum to 180°). Set up the equation: $(4x + 9) + (2x + 33) = 180$ Combine like terms: $6x + 42 = 180$ Subtract 42 from both sides: $6x = 138$ Divide by 6: $x = 23$ Evaluating the angle measures at $x = 23$ gives $4(23) + 9 = 101^\circ$ and $2(23) + 33 = 79^\circ$, which are supplementary ($101^\circ + 79^\circ = 180^\circ$). Common Misconceptions: - Cho.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Two angles form a linear pair. The measures of the angles are $(3x + 10)^\circ$ and $(2x + 20)^\circ$. What is the value of x?

a. 10

b. 12

c. **30** ✓

d. 100

By definition, two angles that form a linear pair are adjacent and supplementary, meaning the sum of their measures is equal to 180°. Set up the algebraic equation: $(3x + 10) + (2x + 20) = 180$ Combine like terms: $5x + 30 = 180$ Subtract 30 from both sides: $5x = 150$ Divide both sides by 5: $x = 30$ Analysis of incorrect choices: - Choice A (10): Results from setting the two expressions equal to each other ($3x + 10 = 2x + 20$), confusing a linear pair with vertical angles. - Choice B (12): Results from s.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Two parallel lines are cut by a transversal. One angle measures 117°. If a second angle forms a pair of alternate exterior angles with it, what is the measure of the second angle?

a. 127°

b. 63°

c. 90°

d. **117°** ✓

Alternate exterior angles formed by a transversal crossing parallel lines are equal. The second angle also measures 117°.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Two parallel lines are cut by a transversal. One angle measures 66°. If a second angle forms a pair of alternate interior angles with it, what is the measure of the second angle?

a. **66°** ✓

b. 76°

c. 114°

d. 90°

Alternate interior angles formed by a transversal crossing parallel lines are equal. The second angle also measures 66°.

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28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Two parallel lines are cut by a transversal. One angle measures 55°. If a second angle forms a pair of corresponding angles with it, what is the measure of the second angle?

a. 65°

b. 90°

c. **55° ✓**

d. 125°

Corresponding angles formed by a transversal crossing parallel lines are equal. The second angle also measures 55°.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Two parallel lines are cut by a transversal. One angle measures 144°. If a second angle forms a pair of vertical angles with it, what is the measure of the second angle?

a. 90°

b. 36°

c. **144° ✓**

d. 154°

Vertical angles formed by a transversal crossing parallel lines are equal. The second angle also measures 144°.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

An angle measures 3 times its supplement. What is the measure of the angle?

a. 120°

b. **135° ✓**

c. 90°

d. 45°

Let the supplement be s. Then angle $= 3s$ and $s + 3s = 180$, so $s = 45^\circ$ and the angle $= 3 \times 45 = 135^\circ$.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

An angle measures 5 times its complement. What is the measure of the angle?

a. 65°

b. 60°

c. **75° ✓**

d. 15°

Let the complement be c. Then angle $= 5c$ and $c + 5c = 90$, so $c = 15^\circ$ and the angle $= 5 \times 15 = 75^\circ$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Two parallel lines are cut by a transversal, forming a pair of corresponding angles $(6x + 1)^\circ$ and $(5x + 6)^\circ$. Find the measure of each angle.

a. 149°

b. **31° ✓**

c. 41°

d. 5°

Corresponding angles are equal: $6x + 1 = 5x + 6$. Solving gives $x = 5$, so each angle $= 6(5) + 1 = 31^\circ$.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Alternate interior angles (parallel lines + transversal) are:

a. Supplementary

b. Complementary

c. **Equal ✓**

d. Adjacent

Applying Angles principles, Equal. Therefore, the verified correct option is Equal.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Two angles are vertical angles. One measures 56°. What is the measure of the other?

a. 124°

b. **56° ✓**

c. 304°

d. 34°

Vertical angles are always congruent, so the other angle also measures 56°.

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Two parallel lines are cut by a transversal. One angle measures 110°. Find the measure of its co-interior (consecutive interior) angle.

a. 110

b. 20

c. 80

d. **70** ✓

Co-interior (consecutive interior) angles sum to 180°: $180 - 110 = 70^\circ$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two angles form a linear pair. One angle measures 161°. What is the measure of the other angle?

a. 71°

b. **19°** ✓

c. 161°

d. 199°

A linear pair is supplementary, summing to 180°. $180 - 161 = 19^\circ$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Two parallel lines are cut by a transversal. Two corresponding angles measure $(3x + 10)^\circ$ and $(2x + 25)^\circ$. Find the measure of the angle.

a. 15

b. **55** ✓

c. 65

d. 125

$3x + 10 = 2x + 25 \Rightarrow x = 15 \Rightarrow$ angle $= 55^\circ$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Foundations & Reasoning

Two angles are complementary. One angle measures 7°. What is the measure of the other angle?

a. **83°** ✓

b. 97°

c. 7°

d. 173°

Complementary angles sum to 90°. $90 - 7 = 83^\circ$.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Coordinate Geometry

Two parallel lines are cut by a transversal. One angle measures 29°. Find the measure of its alternate interior angle.

a. 39

b. 61

c. **29** ✓

d. 151

Alternate interior angles are congruent: 29°.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 11.01 Probability

Two parallel lines are cut by a transversal. One angle measures 33°. Find the measure of its alternate exterior angle.

a. 43

b. **33** ✓

c. 57

d. 147

Alternate exterior angles are congruent: 33°.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Triangles & Congruence

Two parallel lines are cut by a transversal. One angle measures 34°. Find the measure of its corresponding angle.

a. **34** ✓

b. 146

c. 44

d. 56

Corresponding angles are congruent: 34°.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

Two lines intersect, forming vertical angles. One angle measures 28°. Find the measure of its vertical angle.

a. 38

b. 152

c. 62

d. **28** ✓

Vertical angles are congruent: 28°.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

What is the sum of the exterior angles (one at each vertex) of a convex polygon with 9 sides?

a. 1260°

b. 40°

c. 180°

d. **360° ✓**

The sum of the exterior angles of any convex polygon is always 360°, regardless of the number of sides.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

A quadrilateral has three angles measuring 20°, 35°, and 146°. What is the measure of the fourth angle?

a. **159° ✓**

b. 169°

c. 201°

d. 20°

Quadrilateral angles sum to 360°: $360 - 20 - 35 - 146 = 159^\circ$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Circles

How many lines of symmetry does a regular hexagon have?

a. 7

b. **6 ✓**

c. 12

d. 5

A regular n-gon has exactly n lines of symmetry; here $n = 6$.

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